

TC-DBI: A Plug-and-Play Trajectory Confidence-Guided Dynamic Block Inference Strategy for Speech Synthesis with Continuous Block Flow Matching

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Abstract

Block Flow Matching combines block-level autoregression with intra-block parallelism for efficient and high-quality Text-to-Speech (TTS). However, using a fixed block size during inference ignores the non-uniform modeling difficulty of speech, which can cause unstable velocity estimation in challenging regions. This motivates the need for a reliability metric that can guide dynamic adjustment of block size. We propose Trajectory Confidence (TC), a geometric measure derived from trajectory straightness during ODE integration, and introduce TC-Guided Dynamic Block Inference (TC-DBI), a plug-and-play strategy that truncates and regenerates unreliable regions to adapt block size without architectural changes. Experiments show that TC correlates strongly with generation errors, and TC-DBI improves robustness and perceptual quality while maintaining comparable efficiency. Audio samples are available at <https://thuhcsi.github.io/interspeech2026-TC-DBI/>.

Index Terms: Speech Synthesis, Block Flow Matching, Trajectory Confidence

1. Introduction

Modern Text-to-Speech (TTS) systems primarily adopt two paradigms: Autoregressive (AR) and Non-Autoregressive (NAR) models. AR models [1, 2, 3, 4, 5, 6, 7] generate speech sequentially, enabling strong temporal and prosodic dependency modeling but incurring high latency and exposure bias [8, 9, 10]. In contrast, NAR models [11, 12, 13, 14, 15, 16, 17] relax strict sequential conditioning to achieve parallel generation and improved efficiency. However, they often struggle to capture fine-grained temporal dynamics and expressive prosody, partly because parallel generation lacks the instantaneous, incremental context available in AR decoding.

To balance expressiveness and efficiency, block-wise generative modeling [18, 19] has recently emerged as an intermediate paradigm. By partitioning sequences into blocks, these methods maintain long-range contextual coherence across blocks while preserving parallel generation within each block. Among block-wise generative frameworks, continuous Block Flow Matching has recently shown strong performance in speech synthesis due to its continuous latent modeling and stable probability flow formulation. Representative systems such as ARDiT [20] and DragonFM [21] adopt Block Flow Matching for high-fidelity generation. Similar designs have also been explored in music generation (e.g., DiffRhythm2 [22]).

Despite the effectiveness of block-wise modeling, applying it to speech synthesis is inherently constrained by the use of a

fixed block size, which poorly matches the intrinsic characteristics of speech. Speech exhibits highly non-uniform information density. Rapid phonetic transitions and prosodically complex regions demand strong contextual conditioning, whereas steady segments can be generated reliably with limited conditioning. As a result, a fixed block size creates an unfavorable trade-off. Small block size improves conditioning but greatly reduces parallelism and efficiency. While large block size boosts efficiency, it may provide insufficient context for locally difficult regions, which can degrade quality and cause error accumulation. To address this issue, we need a computationally lightweight metric that can quickly assess the reliability of the generated segments and dynamically adjust the block size accordingly.

Prior work [23] leverages token-level probabilities in discrete generative models (e.g., cross-entropy-based confidence) to dynamically control block-wise inference. However, such strategies are difficult to transfer to continuous block-wise models, where ODE-based sampling does not expose an explicit probabilistic confidence signal. Meanwhile, recent works [24, 25] on continuous diffusion and flow-based models explore leveraging sampling-trajectory geometry in probability flow ODEs to improve scheduling strategies. This prompts a natural question: can such trajectory characteristics—readily available during inference—also serve as a training-free signal of local generation reliability?

In this work, we propose Trajectory Confidence (TC), a training-free and computationally lightweight reliability signal. TC directly derives from the sampling trajectory of the probability-flow ODE during inference. It measures the local straightness of the trajectory, and serves as an on-the-fly indicator of generation stability. Building on TC, we introduce Trajectory Confidence-Guided Dynamic Block Inference (TC-DBI), a plug-and-play decoding strategy for continuous Block Flow Matching (and potentially other continuous block-wise ODE samplers). TC-DBI detects unreliable suffix regions within a block and selectively refines them by truncation and regeneration with refreshed contextual conditioning, enabling adaptive block granularity without any model retraining or architectural changes. We apply TC-DBI to our reproduced continuous Block Flow Matching TTS system and demonstrate consistent improvements in robustness and speech quality with minimal inference overhead.

- (1) We propose TC, a training-free and low-cost reliability metric and provide quantitative evidence that TC correlates with generation errors.
- (2) We introduce TC-DBI, a plug-and-play dynamic block inference strategy that adaptively identifies unreliable suffixes and refines them via truncation and regeneration to obtain stronger contextual conditioning in challenging regions.

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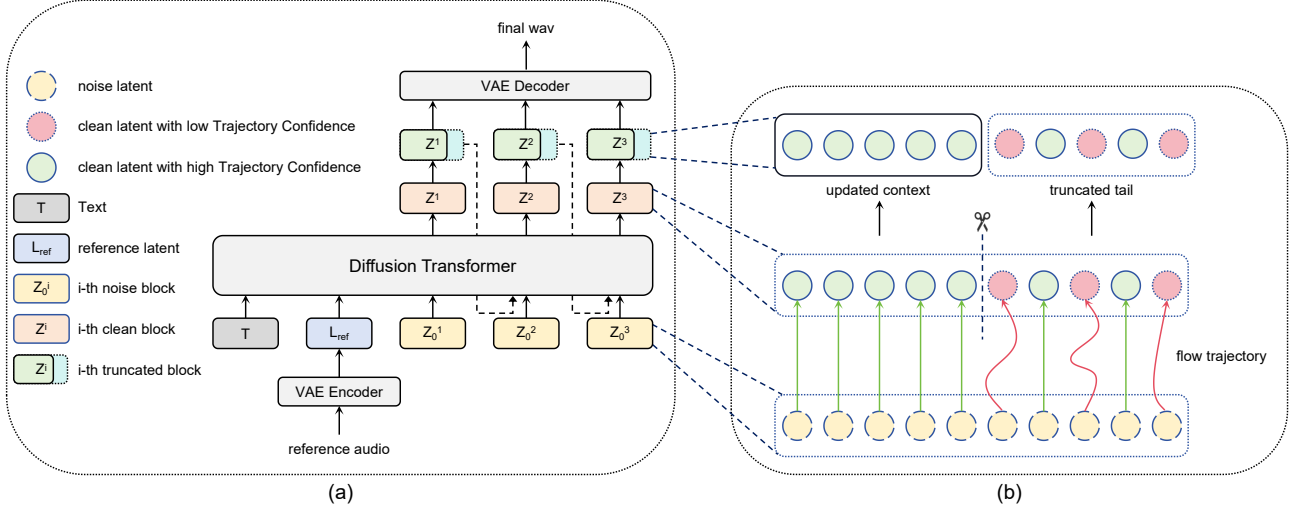


Figure 1: **Overview of Our Method:** (a) Our reproduced Block Flow Matching architecture; (b) Details of TC-DBI.

- (3) We apply TC-DBI to our reproduced Block Flow Matching TTS baseline and substantially improve speech quality and robustness while preserving comparable efficiency.

2. Methodology

2.1. Model Architecture

Fig. 1(a) shows our reproduced Block Flow Matching architecture, which consists of two components:

Speech variational autoencoder (VAE): We employ a pre-trained VAE to compress high-dimensional waveforms into continuous latent variables Z . It operates as a frozen acoustic codec, ensuring high-fidelity reconstruction while reducing the sequence length for efficient generation.

Diffusion Transformer (DiT): The core generator is a DiT designed for block-wise autoregressive modeling. Conditioned on historical blocks, text T , and reference latent L_{ref} , the DiT predicts the flow trajectories for the current block, which are subsequently decoded by the VAE.

2.2. Block Flow Matching

We adopt Conditional Flow Matching (CFM) with Optimal Transport (OT) interpolation to model the generation process in the continuous latent space. Given a target latent sequence Z , we partition it into consecutive fixed-length blocks $Z = \{Z^1, Z^2, \dots, Z^N\}$.

For each block Z^i , we define a conditional flow from Gaussian noise to the data distribution. Specifically, we sample $Z_0^i \sim \mathcal{N}(0, I)$ with the same shape as Z^i , and construct the OT interpolation path $Z_{t^i}^i = (1 - t^i)Z_0^i + t^i Z^i$, where $t^i \in [0, 1]$ denotes the independent noise level for each block i . Under this linear interpolation, the corresponding target velocity field is constant and given by $v^i = Z^i - Z_0^i$.

The model v_θ is trained to regress this velocity field conditioned on historical blocks and external inputs:

$$\mathcal{L} = \frac{1}{N} \sum_{k=1}^N \mathbb{E} \left[\left\| v_\theta(t^i, Z_{t^i}^i, Z^{<i}, T, L_{ref}) - v^i \right\|_2^2 \right] \quad (1)$$

Here, $Z^{<i}$ denotes the historical blocks used for condition-

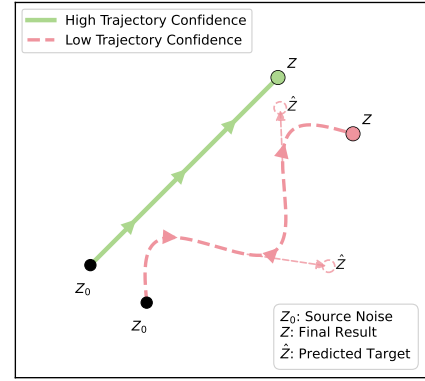


Figure 2: *Paths with different Trajectory Confidence*

ing. During training, it refers to the ground-truth blocks, while during inference it is replaced by the generated blocks $\hat{Z}^{<i}$, ensuring autoregressive dependency at the block level.

During inference, generation proceeds sequentially over blocks. For the i -th block, we initialize $\hat{Z}^i(0) \sim \mathcal{N}(0, I)$ and solve the following conditional ordinary differential equation (ODE):

$$\frac{d\hat{Z}^i(t)}{dt} = v_\theta(t, \hat{Z}^i(t), \hat{Z}^{<i}, T, L_{ref}), \quad t \in [0, 1]. \quad (2)$$

To enable variable-length generation, we append End-of-Prediction (EOP) frames to the acoustic latents and jointly model them under the same flow objective. EOP frames are padded to complete the final block during training, and inference terminates once the EOP signal is detected.

Traditional Block Flow Matching adopts a fixed block length throughout inference, which fails to account for the non-uniform modeling difficulty of speech signals. To address this issue, we propose a training-free and computationally lightweight metric that can quickly assess the reliability of the generated segments and dynamically adjust the block size accordingly, as detailed in the following sections.

2.3. Trajectory Confidence: Definition & Geometric Intuition

To overcome the limitations of fixed block size, we need a metric that assesses generation reliability to dynamically adjust the block size. Under ideal optimal transport interpolation, the probability flow follows a straight path from noise to data; deviations from this straightness can indicate local modeling difficulty and instability in the estimated velocity field. This instability can reflect the model’s predictive confidence. As illustrated in Fig. 2, when the model is certain about the target output, the trajectory remains approximately straight across integration steps. If the model is uncertain about the target, its velocity predictions may become inconsistent across integration steps and lead to curved trajectories and unreliable results.

Motivated by this observation, we quantify Trajectory Confidence using the straightness of the integration path. For the k -th frame within a generated block, assuming the ODE solver performs S steps and produces trajectory points $\{z_k^{(0)}, \dots, z_k^{(S)}\}$, we define the trajectory confidence C_k as:

$$C_k = \frac{\|z_k^{(S)} - z_k^{(0)}\|_2}{\sum_{s=1}^S \|z_k^{(s)} - z_k^{(s-1)}\|_2 + \epsilon}. \quad (3)$$

where the numerator denotes the Euclidean displacement from start to end, and the denominator represents the total path length traversed. ϵ is a small constant introduced for numerical stability. The value of C_k lies in $(0, 1]$, where a value closer to 1 indicates a straighter trajectory and higher confidence.

2.4. Trajectory Confidence-Guided Dynamic Block Inference (TC-DBI)

Building upon the proposed Trajectory Confidence metric, we propose TC-DBI that adaptively adjusts the effective generation length according to local modeling difficulty.

As illustrated in Fig. 1(b), at each generation stage, the model first performs parallel generation of a candidate block with a maximum length L_{max} , following the standard fixed-block inference procedure. After ODE integration, trajectory confidence is computed for each frame using Eq. 3, yielding a confidence sequence $\mathcal{C} = [C_1, \dots, C_{L_{max}}]$.

Rather than accepting the entire block, we determine the longest reliable prefix based on a predefined confidence threshold τ . Specifically, we define the truncation boundary as:

$$k^* = \begin{cases} \max(2, \min\{k \mid C_k < \tau\}), & \text{if } \exists k \text{ s.t. } C_k < \tau, \\ L_{max} + 1, & \text{otherwise.} \end{cases} \quad (4)$$

The prefix $\{1, \dots, k^* - 1\}$ is regarded as reliable and appended to the generated sequence, while the remaining frames are truncated. Notably, Eq. 4 strictly enforces a minimum retention of one frame per iteration. This prevents a potential infinite loop: if all frames were truncated, the unchanged conditioning context may cause the model to repeatedly generate the identical trajectory. Generation then resumes from position k^* , using the retained prefix as updated conditioning context.

This Trajectory Confidence guided mechanism dynamically modulates the effective block size during inference. In regions where the flow trajectory remains approximately straight, large parallel steps are preserved for efficiency. In contrast, when trajectory deviations indicate local uncertainty, the block size is automatically reduced, preventing unreliable predictions from propagating across subsequent blocks.

3. Experiments

3.1. Experimental Setup

Our reproduced Block Flow Matching (BFM) is built upon a Diffusion Transformer (DiT) with 22 layers, hidden dimension 1024, and 16 attention heads. Block-wise autoregressive generation is performed with a fixed block size of 32 during training. For acoustic representation, we adopt the open-source VoxCPM [26] VAE, which compresses 44.1 kHz waveforms into 25 Hz continuous latent sequences, serving as a frozen neural codec. Our model is trained on approximately 100k hours of speech data from the Chinese and English subsets of Emilia. Training is conducted on 16 NVIDIA A100 (80G) GPUs for approximately two weeks. During inference, we perform ODE integration with 32 steps. For TC-DBI, the trajectory confidence threshold is fixed to $\tau = 0.75$ to determine the truncation boundary.

3.2. Experiment Results

3.2.1. Main Results

Table 1 reports the main results on the Seed-eval benchmark. Since no publicly available BFM model matches our training scale, we reproduced a Block Flow Matching model (BFM) as a baseline. For a comprehensive comparison, we also include CosyVoice2, MaskGCT, and F5-TTS, covering AR and NAR paradigms.

We evaluate performance using three metrics. Word Error Rate (WER, $\downarrow$) measures intelligibility via Whisper-large-v3 for English and Paraformer-zh for Mandarin and reflects generation robustness. Speaker Similarity (SIM, $\uparrow$) calculates the cosine similarity of speaker embeddings extracted by a fine-tuned WavLM-large model, which quantifies speaker consistency between synthesized and reference audio. Naturalness MOS (N-MOS, $\uparrow$) evaluates perceptual quality, rated by 15 experienced listeners on 10 utterances per language.

As shown in Table 1, BFM with TC-DBI demonstrates the most balanced performance across both test sets. It attains the lowest WER on seed-test-en (1.798%) and the second-best WER on seed-test-zh (1.568%), close to CosyVoice2 (1.504%), indicating high content fidelity while maintaining strong perceptual quality. Notably, our method delivers the highest N-MOS on seed-test-zh/en (3.809/3.973), surpassing the best baselines by consistent margin, demonstrating improved realism and prosody that generalizes across languages. Speaker similarity is comparable to the strongest systems, suggesting that the improvements are not driven by similarity optimization but by enhanced perceptual quality. Finally, removing TC-DBI degrades WER (zh: 1.568% $\rightarrow$ 1.628%; en: 1.798% $\rightarrow$ 1.921%) and N-MOS (zh: 3.809 $\rightarrow$ 3.673; en: 3.973 $\rightarrow$ 3.727) with minimal SIM change, confirming the effectiveness of TC-DBI in improving robustness and perceptual quality.

3.2.2. Validity of Trajectory Confidence

Figure 3 presents a combined error density analysis on the Chinese and English subsets of Seed-eval. For each utterance, we use the minimum Trajectory Confidence as its sentence-level reliability score. We progressively filter samples whose score falls below a given threshold and compute the percentage of erroneous samples (WER > 0) within the filtered subset. The red curve shows the error probability of the selected subset, while the gray dashed line indicates the global baseline error rate (21.36%). The blue bars (log scale) denote the number of filtered samples at each threshold. A clear monotonic trend

Table 1: Main evaluation results on the Seed-eval dataset in terms of WER ($\downarrow$), SIM ($\uparrow$), and N-MOS ($\uparrow$)

Method	seed-test-zh			seed-test-en		
	WER $\downarrow$	SIM $\uparrow$	N-MOS $\uparrow$	WER $\downarrow$	SIM $\uparrow$	N-MOS $\uparrow$
CosyVoice2 [6]	1.504%	0.752	3.391 ± 0.149	2.634%	0.658	3.700 ± 0.129
F5-TTS [16]	1.640%	0.760	3.509 ± 0.137	<u>1.830%</u>	0.672	3.545 ± 0.144
MaskGCT [17]	2.330%	0.773	3.191 ± 0.148	3.936%	0.715	3.409 ± 0.136
BFM (ours)	1.628%	0.764	3.673 ± 0.123	1.921%	0.676	3.727 ± 0.145
BFM with TC-DBI (ours)	<u>1.568%</u>	0.753	3.809 ± 0.157	1.798%	0.676	3.973 ± 0.134

emerges: as the confidence threshold becomes stricter, the error density increases significantly. For example, when the threshold is set to 0.65, the error rate rises to 35.0%, corresponding to a 1.64 $\times$ increase over the global baseline. As the threshold relaxes toward 0.85 and above, the error rate gradually converges to the baseline level. This behavior demonstrates that sentences with lower Trajectory Confidence are substantially more likely to exhibit recognition errors. These results provide strong statistical evidence that Trajectory Confidence serves as a reliable, training-free quality estimator, justifying its use for adaptive block inference.

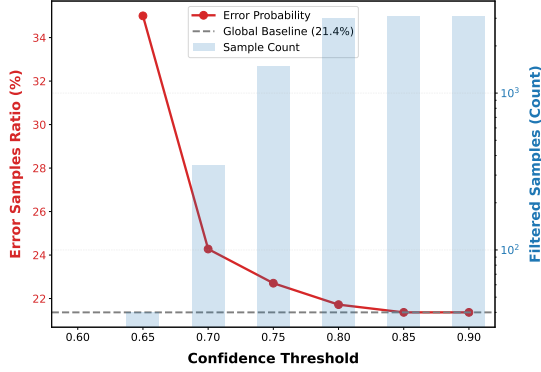


Figure 3: Error enrichment analysis under varying Trajectory Confidence thresholds

3.2.3. Ablation Study: Impact of Confidence Threshold τ

To further examine how trajectory confidence translates into inference gains, we evaluate TC-DBI under different thresholds τ (Table 2). Across all tested thresholds, TC-DBI consistently improves WER over the Fixed-Block baseline of our reproduced BFM. At $\tau = 0.65$, only very low-confidence segments are truncated, yielding limited WER reduction due to their low frequency. Increasing τ to 0.75 allows TC-DBI to cover a broader set of moderately low-confidence segments, resulting in the best WER on both Seed-ZH (1.568%) and Seed-EN (1.798%) while incurring only a modest 1.08 $\times$ relative RTF. Further increasing τ to 0.80 triggers more aggressive truncation, which raises RTF to 1.63 $\times$ with diminishing returns. These results demonstrate that trajectory confidence effectively identifies high-risk regions, and that balancing error density and occurrence frequency is crucial for optimal performance.

3.2.4. Trajectory Confidence Improvement Analysis

To further evaluate the effectiveness of dynamic regeneration, we analyze the confidence distribution of frames that were originally below the threshold ($\tau = 0.75$) after backtracking and regeneration. As shown in Fig. 4, more than 90% of these frames

Table 2: Ablation study on confidence threshold τ . “Relative RTF” is normalized to the Fixed-Block baseline (lower is faster).

Method	RTF	Seed-ZH		Seed-EN	
		WER $\downarrow$	SIM $\uparrow$	WER $\downarrow$	SIM $\uparrow$
Fixed-Block ($\tau = 0$)	1.00 $\times$	1.628%	0.764	1.921%	0.676
TC-DBI ($\tau = 0.65$)	1.00$\times$	1.621%	0.764	1.810%	0.676
TC-DBI ($\tau = 0.70$)	1.01 $\times$	1.619%	0.753	1.839%	0.676
TC-DBI ($\tau = 0.75$)	1.08 $\times$	1.568%	0.753	1.798%	0.676
TC-DBI ($\tau = 0.80$)	1.63 $\times$	1.608%	0.754	1.876%	0.675

are promoted above the predefined threshold, and nearly half exceed 0.85, indicating a substantial recovery in trajectory reliability. This shift suggests that regeneration successfully transforms previously uncertain regions into high-confidence predictions. The improvement can be attributed to strengthened contextual conditioning: once unreliable suffixes are removed, generation resumes from a confirmed reliable prefix, providing more stable temporal context and reducing trajectory deviation. As a result, the model produces more deterministic and trustworthy outputs.

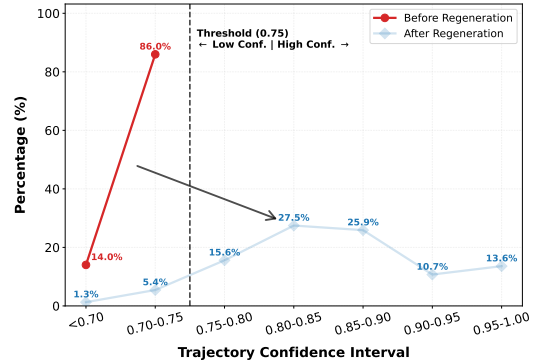


Figure 4: Trajectory Confidence improvement analysis

4. Conclusion

We proposed Trajectory Confidence (TC) as a training-free and computationally lightweight reliability metric for continuous Block Flow Matching in text-to-speech synthesis. Building on TC, we introduce TC-DBI, a plug-and-play decoding strategy that adaptively adjusts block-wise generation by truncating low-confidence regions and regenerating uncertain segments, requiring no architectural changes or retraining. Experiments on bilingual Seed-eval benchmarks demonstrate that TC-DBI effectively identifies high-risk regions, improves robustness, and enhances perceptual quality while maintaining efficiency and speaker similarity.

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6. Generative AI Use Disclosure

During the preparation of this manuscript, the authors used generative AI tools exclusively for the purpose of language editing and manuscript polishing to improve readability. These tools were not used to generate any core scientific ideas, experimental data, or technical contributions. All authors have thoroughly reviewed and approved the final version of the manuscript, and assume full responsibility for the integrity and entirety of its content.

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